

RAJAT SAVDEKAR

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EDUCATION

University of Southern California, Los Angeles	Aug 2024 – May 2026
Master of Science, Computer Science	GPA: 3.6
University of Mumbai, Mumbai	Aug 2019 – May 2023
Bachelor of Engineering, Computer Engineering	GPA: 3.7

WORK EXPERIENCE

USC Summer Programs	May 2025 – Jul 2025
Teaching Assistant	Los Angeles, CA
Led 2 international teams of 15+ students through Python, Git, and RESTful API development, guiding them to build and present project websites using Google Sites.	
Jalgaon Fruit Sales Cooperative Ltd – Agritech Supply Chain Platform	May 2023 – May 2024
Software Engineer	Jalgaon, India
- Migrated 150 daily users off manual logs by building a JavaScript procurement platform backed by a RESTful API, sustaining 99.9% uptime across complex inventory transactions. - Cut average query time 40% on high-frequency lookups by decomposing an undocumented legacy monolith and redesigning the MySQL schema and indexing strategy. - Reduced deployment cycle time 30% and hardened data flows between microservices by introducing Docker-based CI/CD with Git-managed releases and pytest unit and integration tests.	

SELECTED PROJECTS & OPEN SOURCE

Celestia – Meta Quest VR Stargazing Experience	Jun 2026
Unity, C#, AWS, MongoDB	[Demo]
- Built a Meta Quest 3 VR astronomy explorer grounding an LLM with real-time orbital data by engineering a Python RAG pipeline and RESTful API indexing 10,000+ celestial bodies. - Sustained 90 FPS for 6DoF planetary models by optimizing CI/CD-deployed Unity rendering pipelines, extended into touch-optimized Android (Flutter) and iOS (Swift) apps.	
MockPad – Shared Code Editor	May 2026
React, Node.js, Yjs, MongoDB	[Live] [GitHub]
- Built a full-stack real-time collaborative code editor, architecting around Yjs CRDTs over a custom WebSocket layer for 50+ simultaneous clients with conflict-free concurrent edits. - Integrated OpenAI Whisper for async transcription into a three-timeline event-sourced log (code, audio, captions), deployed on Vercel/DigitalOcean and backed by MongoDB for full session replay.	
RDX Dev Creator Lab – Developer Visualization Platform	
Python, React, Manim, Remotion, Godot, Three.js	[GitHub]
- Built an open-source developer visualization platform that programmatically renders system architecture diagrams, algorithms, and interactive technical documentation. - Automated speech alignment, rendering, and asset-generation workflows with reusable Python tooling around Whisper, FFmpeg, and CLI scripting.	
SplitIt – Expense Tracker	Jul 2026
React, Node.js, Express.js, MongoDB, JWT	[GitHub]
- Built a full-stack expense tracker with secure JWT authentication, protected REST APIs, and friend/group management for collaborative expense CRUD operations. - Shipped dynamic React dashboards with real-time balance visualization and activity feeds across group and expense workflows.	

HONORS & PUBLICATIONS

Aura – visionOS Accessibility Assistant. 2nd-place, 24-hour hackathon-winning application built with Swift, SwiftUI, and RealityKit, delivering fully on-device live captions and environmental sound awareness.	[GitHub]
Research Publications. Published multiple peer-reviewed papers with 10+ citations across IEEE and Springer, covering computer vision, image processing, and machine learning for reproducible scientific research.	[GitHub]

SKILLS

Languages: Python, Java, JavaScript, TypeScript, Swift, C++, C#, SQL
Frameworks & Technologies: React, Next.js, Node.js, Flask, REST APIs, GraphQL, Microservices
AI & Data: LLMs, RAG, PostgreSQL, MySQL, MongoDB
Cloud & Tools: AWS, Docker, Git, CI/CD